

Project Proposal: Gurugram Cyber Scam Map & Complaint Analytics Dashboard

1. Overview & Problem Statement

Cybercrime in India has grown rapidly, with NCRB data showing over 86,000 recorded cybercrime cases in 2023, a rise of more than 30% compared to 2022, and nearly 69% of these cases are related to fraud. Haryana has seen a steep increase in cybercrime complaints over the last five years, with daily complaint volumes in the hundreds and substantial financial losses reported. Gurugram, in particular, has emerged as a major cybercrime hotspot in the state, registering tens of thousands of complaints annually and losses that have surged from a few crores to over ₹100 crore in a few years.^{[1][2][3][4][5][6][^7]}

Despite the availability of the National Cybercrime Reporting Portal (cybercrime.gov.in) and the 1930 helpline for reporting cyber frauds, investigators still face challenges in quickly understanding local patterns of crime, such as which scam types dominate in specific areas, which demographics are most affected, and how complaints evolve over time. Current reporting systems primarily capture cases and complaints, but actionable, location-focused visual intelligence for a single district like Gurugram is limited.^{[8][9][10][11]}

This project proposes a **Gurugram Cyber Scam Map & Complaint Analytics Dashboard**: a fullstack web application tailored for Gurugram Police that visualizes cybercrime complaint data on an interactive map and through analytical charts, enabling faster insights into fraud hotspots, victim profiles, and scam trends.

2. Project Objectives

- Provide a single integrated platform that combines a scam heatmap of Gurugram with a complaint analytics dashboard.
- Help investigators and leadership quickly answer questions like:
 - “Which police station areas report the most UPI frauds this month?”
 - “Where are sextortion and work-from-home frauds concentrated?”
 - “Which age group and gender are most targeted by specific scam types?”
- Support data-driven awareness campaigns by identifying the most affected localities and common modus operandi.
- Offer a role-based interface so public-facing views show high-level trends and safety tips, while investigator views show deeper, operational details.

3. Target Users

- **Primary:** Gurugram Cyber Police and district-level investigators.
- **Secondary:** Senior police leadership for decision-making and review.
- **Tertiary (optional public view):** Citizens, for awareness on scam hotspots and prevalent fraud types in their locality.

4. System Overview & Architecture (High Level)

4.1 Key Modules

- **Data Ingestion & Management Module**
 - Import complaint data (CSV upload) with fields like date, scam type, location (PS/beat/ward or pincode), amount lost, victim age group, gender, and channel used.
 - Manual case entry form for new complaints during demos.
- **Analytics Engine**
 - Aggregates complaints by scam type, time period, geography, and demographics.
 - Computes simple metrics such as total cases, total amount lost, recovery amount (if available), and a configurable risk score per area.
- **Scam Map Module**
 - Displays Gurugram map with color-coded regions (heatmap) by complaint density, filterable by scam type and time window.
- **Dashboard & Visualization Module**
 - Charts for trends over months/years, age group distribution, gender breakdown, and channel breakdown (UPI, cards, social media, OTP, etc.).
- **Role-Based Access & Views**
 - Investigator/Admin view: full data, download/export, detailed filters.
 - Public/awareness view (optional): anonymized, high-level statistics and safety tips.

4.2 Suggested Tech Stack

- **Frontend:** React.js, Tailwind CSS, charting library (e.g., Chart.js or Recharts), map library (e.g., Leaflet or Mapbox GL JS).
- **Backend:** Node.js + Express (or FastAPI in Python) exposing REST APIs.
- **Database:** PostgreSQL or MongoDB for complaint records and metadata.
- **Real-Time Layer (Optional):** WebSocket or Server-Sent Events for live dashboard updates when new complaints are added.
- **Deployment:** Frontend on Netlify/Vercel, backend on Render/Railway; database on cloud (e.g., Railway PostgreSQL).

5. Core Features (MVP)

5.1 Scam Heatmap of Gurugram

- Interactive map of Gurugram showing heat intensity by complaint count.
- Filters:
 - Scam type (UPI fraud, QR fraud, task fraud/work-from-home scam, sextortion, investment scam, job scam, fake KYC, electricity bill scam, etc.).^{[4][6][^7]}
 - Time range (last 7 days, 30 days, quarter, year).
 - Amount range (e.g., micro-frauds vs high-value cases).
- Tooltip for each area showing:
 - Number of complaints.
 - Top 3 scam types in that area.
 - Optional awareness snippet: a short description of common modus operandi in that locality.

5.2 Complaint Analytics Dashboard

- **Overview Cards:** Total complaints, total reported loss, average loss per complaint, and (if provided) recovered amount.
- **Trend Graphs:**
 - Time-series chart of complaints over months/years.
 - Time-series of total amount lost.
- **Demographic Charts:**
 - Pie/bar charts by age group and gender.
 - Charts by channel: mobile apps, UPI, social media, work-from-home/task scams, OTP fraud, etc..^{[7][4]}
- **Fraud Type Distribution:**
 - Bar chart showing counts by scam type, reflecting that fraud-related motives dominate cybercrime at the national level.^{[5][1]}

5.3 Complaint Management & Data Entry

- Internal form for adding a new complaint record with fields like:
 - Date/time of complaint.
 - Scam type.
 - Location (police station, locality, or pincode).
 - Channel used (UPI, card, social media, etc.).
 - Amount reported.

- Victim age range and gender (optional, anonymized).
- Bulk CSV upload feature for importing historical data (e.g., synthetic demo data or anonymized aggregates from internal systems).

5.4 Role-Based Views

- **Investigator/Admin View:**
 - Full dashboard plus case table with advanced filters (by PS, type, date, amount, age group).
 - Export options (CSV/Excel) for sharing in briefs and reports.
 - Access to detailed heatmap and metrics.
- **Public View (Optional):**
 - Only aggregated, anonymized statistics and heatmap.
 - Prominent display of 1930 helpline and cybercrime.gov.in portal details to encourage reporting.^{[9][12][10][8]}

6. Advanced / Differentiating Features

6.1 Area Risk Scoring

- Simple, explainable risk score per locality based on:
 - Complaint volume + recent growth.
 - Total amount lost.
 - Diversity of scam types (areas with many different scams treated as higher risk).
- Risk levels (Low/Medium/High) visually indicated on the map and in a ranked list.

6.2 Scam Playbooks & Awareness Snippets

- For each scam type (e.g., task fraud, work-from-home scam, sextortion, fake KYC), attach a short "playbook" describing:
 - Typical approach message.
 - Red flags to watch out for.
 - Steps to take immediately (call 1930, preserve evidence, report on cybercrime.gov.in, etc.).^{[11][4][^9]}
- On the public view, when a user clicks an area or scam type, show localized awareness content.

6.3 Live Updates (Real-Time Mode)

- Use WebSockets/Server-Sent Events to broadcast new complaints added via the internal form so that the dashboard updates without refresh.
- Useful in live monitoring scenarios (e.g., during a specific operation or campaign period).

6.4 Integration with External Data (Future-ready)

- Design API endpoints and database schema to accommodate integration with:
 - NCRP exports (if format is shared in future).
 - Additional datasets from NCRB or I4C for cross-validation of state- and national-level trends.^{[13][1][^11]}

7. Future Enhancements & Roadmap

The project can be extended in multiple directions beyond the internship timeline:

- **Predictive Analytics (Phase 2):**
 - Use basic time-series forecasting to flag weeks/months where certain scam types are likely to spike.
 - Identify “emerging scam types” by looking for sudden increases in specific categories.
- **Automated Alerts:**
 - Configurable thresholds to send an alert (email/notification) when a locality crosses a set limit of complaints or total amount lost.
- **Deeper Citizen-Facing Portal:**
 - Allow citizens to search their pincode/locality and see anonymized stats plus tailored safety advice.
 - Embed multilingual awareness videos and infographics aligned with national campaigns.
- **Integration with Cyber Awareness Chatbot:**
 - Combine with an AI-powered cyber safety assistant so that citizens can chat about a suspicious message and get both advice and context on whether similar scams are trending in their area.
- **Case Workflow Hooks (Long-Term):**
 - Add fields to link dashboard records with FIR numbers and investigation status.
 - Provide basic workflow stats: number of complaints converted to FIRs, arrests, and recovery ratios.

8. Uniqueness & Impact

- **Combines two flagship ideas into one system:** Instead of choosing either a complaint dashboard or a scam map, this project fuses both into a single, cohesive platform tailored to Gurugram, aligning strongly with the GPCSSI flagship theme.
- **Local, high-resolution intelligence:** While national and state-level cybercrime trends are published by NCRB and other bodies, district-level, map-based visualization focused specifically on Gurugram is far less common, despite the district's known cybercrime burden and high complaint counts.^{[2][6][4][7]}
- **Direct support to existing mechanisms:** The tool complements—not replaces—NCRP and helpline 1930, by turning raw complaints into operational intelligence and awareness content.^{[10][8][9][11]}
- **Built on your proven strengths:** The project leverages frontend dashboard design, performance optimization, real-time features, and API integration skills demonstrated in previous hackathon-winning and finalist projects, making it both ambitious and realistically deliverable within 4–6 weeks.^[^14]
- **Showcase-ready:** Visually rich dashboards, interactive maps, and localized scam playbooks make it ideal for live demo during the valedictory ceremony, with a clear storyline: “This is how Gurugram Police can see, understand, and respond to cyber fraud patterns in their own city.”

9. Proposed Timeline (4–6 Weeks)

- **Week 1:** Finalize data model, design low-fidelity UI/UX, set up project structure and database.
- **Week 2:** Implement backend APIs for complaints, aggregation, and filters; seed database with synthetic/demo data.
- **Week 3:** Build core frontend dashboard (overview cards, charts, filters) and wire up API integration.
- **Week 4:** Implement scam heatmap, tooltips, and role-based views (investigator vs public).
- **Week 5:** Add advanced features (risk scoring, awareness snippets, CSV export, optional real-time updates).
- **Week 6:** Testing, performance tuning, polishing UI, documentation, and final demo preparation.

This proposal is designed so that the initial MVP is feasible within the internship while leaving ample scope for future enhancements and deeper integration with Gurugram Police workflows.

References

1. [Crime in India 2023 - NCRB Report Highlights Surge in Cybercrime ...](#) - The NCRB has released its Crime in India 2023 report, highlighting a significant increase in cybercr...
2. [Cybercrime on rise in Gurugram, over 19K complaints in 6 ...](#) - Sanjay Yadav Gurugram, July 11 Despite awareness campaigns, cybercrime is on the rise in Gurugram. M...
3. [NCRB data shows rise in cybercrimes using mobile phones and ...](#) - The total recorded cybercrime cases in India in 2021 were over 52,000, which rose to over 86,000 in ...
4. [79 fall victim to cybercrime in Gurugram daily - The Tribune](#) - Sanjay Yadav Gurugram, July 17 Cybercrime cases are on the rise as more than 79 people in Gurugram f...
5. [\[Solved\] According to the Crime in India 2023 report released by the](#) - India recorded 62.4 lakh cognizable crimes in 2023, marking a 7.2% rise from 2022. Cybercrimes surge...
6. [In 4 years, Ggm cyber fraud losses surge by 12 times](#) - According to Gurugram police, the amount of money that cybercriminals have defrauded from the reside...
7. [In 5 Yrs, Hry Sees 10-fold Increase In Cybercrimes | Chandigarh News - Times of India](#) - Chandigarh: Haryana has registered nearly 10-fold increase in the registration of complaints of cybe...
8. [National Cybercrime Reporting Portal - Wikipedia](#)
9. [Cyber Frauds Helpline](#) - The Ministry of Home Affairs has operationalized a toll-free Helpline number '1930' (earlier '155260
10. ['National Cyber Crime Reporting Portal' enables public to ...](#)
11. [National Cybercrime Reporting Portal \[NCRP\]](#) - Ministry/Department
12. [cybercrime helpline number 1930. - Cyber Crime Portal](#) - Ministry/Department
13. [Crime in India - 2023 - Dataset - CKAN](#) - Crime in India report 2023 published by the National Crime Records Bureau (NCRB) ... Cyber Crimes (C...
14. [Abhay_res_latest.pdf](#)